Transferability audit of data-center cooling life-cycle assessment considering lifecycle electricity, functional equivalence, and rank robustness

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Abstract

Published data-center cooling life-cycle assessments (LCAs) are difficult to transfer because electricity boundaries, climate methods, useful-computation equivalence, and foreground inventories differ. This Technology Assessment uses OpenDC-LCA, an open-source evidence-audit and comparison-validation tool, to test which conclusions from the released Microsoft/WSP comparison remain supportable. We reconstructed 24 totals, harmonized nine U.S. Environmental Protection Agency Emissions & Generation Resource Integrated Database (eGRID) releases to the Intergovernmental Panel on Climate Change Fifth Assessment Report 100-year global warming potential (AR5 GWP100), and calculated 71 non-residual 2023 Federal

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LCA Commons electricity-consumption systems. The article reports AR5 GWP100, but its archived workbook places the numeric electricity endpoints in 100-year global temperature-change-potential (GTP100) cells while the corresponding GWP100 cells are blank and the comparative use-phase formulas reference those blank cells. We therefore retain cooling totals only as numerical transferability screens. The harmonized national direct-generation factor declined 32.46%, from 517.73 to 349.67 kg carbon dioxide equivalent per megawatt-hour between 2012 and 2023. The partial linked national consumption system yielded 422.93 kg carbon dioxide equivalent per megawatt-hour but retained 287 unlinked technosphere inputs. Under equal relative bounds, the first-rank screen failed at a 1.32% all-block half-width. Service-equivalence and use-phase thresholds were 2.64% and 2.99%, compared with 22.15% for embodied burden. This result reverses an earlier attribution of rank sensitivity to embodied variation. The audit supports the decline in generation-related greenhouse-gas intensity and identifies decision-critical data, but it does not identify a universally preferable cooling architecture.
Keywords: data center, life-cycle assessment, energy efficiency, lifecycle electricity, functional equivalence, sensitivity analysis, open-source software

1. Introduction

1.1. Cooling-system lifecycle context

Data centers couple rapidly changing computing hardware to long-lived buildings, electrical systems, cooling plants, and regional energy and water systems. Global data-center electricity estimates remain uncertain, but the scale and growth of cloud and artificial intelligence (AI) workloads make energy efficiency and electricity-supply emissions simultaneous design constraints [1, 2]. At the equipment level, increasing processor heat flux and rack density are driving a transition from room air cooling toward rear-door heat exchangers, direct-to-chip cold plates, and single- or two-phase immersion [3, 4, 5].

These cooling architectures alter more than the heat exchanger because they also change server fans, pumps, coolant distribution units, heat-rejection temperature, fluid inventory, rack or tank construction, packing density, repair procedures, and sometimes achievable clock rate. System experiments have reported materially different coefficient of performance and cost behavior for single- and two-phase immersion over load [6], while recent single-phase tests show that flow direction, coolant viscosity, and water temperature affect chip temperature, thermal resistance, and power usage effectiveness (PUE) [7]. Two-phase experiments similarly show that boiling, pressure stability, and condenser performance must be evaluated at server and system levels [8]. A comparison at equal rack count, equal IT electricity, or equal nameplate capacity risks comparing different computational services.

PUE remains useful for facility energy management, but it is not a lifecycle functional unit and does not represent hardware production, cooling-fluid loss, replacement, water scarcity, or useful computation. Cooling selection is an energy-system decision with lifecycle consequences, not a single-metric efficiency contest.

1.2. Data-center LCA literature

Early environmental assessments showed that operating electricity dominates many data-center footprints and that PUE alone can transfer burdens outside the measured facility boundary [9]. U.S. studies subsequently linked data centers to spatially varying electricity and water burdens [10, 11]. At the product scale, Isler-Kaya and Karaosmanoglu collected manufacturing inventory for a cooling device and evaluated electricity and waste scenarios [12]. Wenzel and Radgen compared energy, material-flow, exergy, and life-cycle assessment methods for a wet cooling tower, illustrating that method choice changes the question answered [13].

Alissa et al. published the most detailed public hyperscale comparison of air, cold-plate, single-phase immersion, and two-phase immersion [14]. Their cradle-to-grave model included buildings, support equipment, servers, racks or tanks, cables, cooling fluids, use-phase electricity and water, replacement, and end of life. The use of Vcore-year as the functional unit was a major advance because it allowed packing density and cooling-enabled computation to enter the comparison. The accompanying archive includes normalized contributions, equations, uncertainty material, and a pedigree assessment [15]. It does not, however, disclose every licensed background process, architecture-specific bill of quantities, or a transferable mapping from electricity to workload performance.

Recent work has moved closer to architecture- and service-level assessment. d'Orgeval et al. modeled complete data-center configurations and reported performance-normalized results for GPU-based subsets [16]. Wu et al. released and validated a cooling-plant virtual testbed against three months of operating data, with 7.62% mean absolute percentage error for power [17]. These studies provide performance and architecture evidence that a secondary normalized table cannot supply. ITU-T L.1410 likewise treats the functional unit and the rules for comparative ICT analysis as explicit study choices [18].

Other LCA work confirms that changes in electricity-generation emissions and cooling efficiency are complementary but can shift impacts among categories [19]. Manufacturer and ICT studies also show wide dispersion in embodied server impacts [20, 21]. Together, the performance and manufacturer evidence leaves a specific transferability gap because available primary studies do not provide a reusable audit for determining whether a published cooling comparison can be moved to a different electricity system without changing its method, boundary, or computational service.

1.3. Temporal and geographic electricity accounting

Electricity is not a scalar independent of method. Attributional LCA usually requires a consumption mix with upstream generation, transmission, and other indirect processes, whereas eGRID state total-output rates describe direct emissions from in-state generation. Location-based, market-based, and marginal factors answer different questions. Combining location- and market-based accounting inconsistently can double count renewable attributes [22, 23]. Average rates can also fail to represent the effect of workload shifting. Dandres et al. found that average and marginal electricity signals can lead to different real-time data-center decisions [24].

Time affects i) cooling performance across load and weather, ii) hourly and long-term grid composition, iii) discrete equipment replacement, and iv) background technology. Dynamic-LCA reviews distinguish temporal inventory, characterization, and prospective background change [25, 26]. Open tools such as Temporalis and, more recently, bw_timex show how temporal information can be propagated through process networks [27, 28, 29]. For cooling, hourly operation is necessary but insufficient. The grid method, geography, and upstream boundary must also be aligned. A weather file cannot repair an incompatible electricity inventory.

1.4. Data quality, uncertainty, and decision value

LCA uncertainty arises from parameters, scenarios, model form, allocation, system boundary, and incomplete knowledge. Monte Carlo propagation is useful when distributions and correlations are supportable. It becomes misleading when arbitrary ranges are presented as confidence. Correlation can materially change propagated uncertainty and cannot be inferred from marginal ranges alone [30]. Reviews recommend separating variability from uncertainty and using sensitivity analysis to identify influential assumptions [31]. Pedigree matrices can communicate reliability, temporal, geographic, and technological representativeness, but the conversion from qualitative scores to uncertainty is itself a modeling choice [32, 33].

EPA guidance treats reproducible documentation at flow and process levels as part of data-quality assessment [33]. Value-of-information methods extend this assessment by prioritizing data whose resolution is expected to change a decision rather than simply improving every weak input [34]. For electronics-cooling laboratories, this distinction separates a poor-pedigree parameter with negligible decision leverage from a small PUE difference that may decide between nearly tied lifecycle alternatives.

ISO 14040 and ISO 14044 define the LCA framework and requirements, whereas ISO 14071 (2024) specifies the critical-review process and reviewer competencies needed for stronger claims, with particular relevance to comparative assertions intended for public disclosure [35, 36]. Code tests, data checks, and coauthor review can strengthen traceability, but none is a substitute for an independent critical review.

1.5. Research gap and contributions

General engines such as openLCA and Brightway calculate process networks, exchange inventories, and apply life-cycle impact assessment (LCIA) methods [37, 38]. They do not determine whether cooling alternatives deliver equivalent computation, whether a performance map covers the operating domain, whether a grid substitution preserves the original electricity boundary, or whether secondary screening results support a public comparative assertion. Conversely, cooling-plant models can validate operational power without closing lifecycle inventories or comparative-review requirements [17].

Four gaps remain at the intersection of cooling engineering and LCA. First, published normalized totals cannot be transparently updated when foreground quantities and licensed process mappings are incomplete. Second, common-unit electricity factors can retain incompatible characterization methods and system boundaries. Third, qualitative data-quality scores do not directly show which measurement could change the technology decision. Fourth, software often produces a number without preserving whether it is reconstructed, screened, empirically supported, or suitable for a comparative claim.

This technology assessment addresses the transferability of the released

Microsoft/WSP case through five linked contributions. First, it reconstructs 24 released totals and audits the workbook cells to identify an unresolved GTP100/GWP100 inconsistency in the public archive. Second, it harmonizes nine eGRID releases from gas-specific emissions. Third, it calculates 71 Federal LCA Commons 2023 consumption systems with an openLCA-compatible JSON-LD workflow and reports process links, numerical residuals, and unlinked technosphere cutoffs rather than hiding them. Fourth, it derives exact, equal-width, deterministic rank-robustness certificates that distinguish the effects of grid, use phase, embodied burden, and service equivalence without assigning unsupported probabilities. Finally, it provides the open-source OpenDC-LCA comparison validator, which blocks incompatible functional units, boundaries, electricity accounting, LCIA methods, or missing field-level lineage while stating that metadata checks are not physical validation.

Together, the reconstruction addresses the missing public arithmetic in gap 1. The harmonization and linked-system calculations expose the method and boundary conflicts in gap 2. The rank-robustness certificates identify the measurements with decision leverage in gap 3. The OpenDC-LCA validator preserves the evidence class needed to prevent gap 4. OpenDC-LCA is therefore the reproducible implementation of this assessment layer, not a replacement for the process-network engines openLCA or Brightway.

The aim is not to reconstruct licensed background inventories or declare a winning cooling technology. It is to determine which claims survive a transferability audit and which require primary measurement, complete inventories, or external critical review.

2. Framework and methods

2.1. Evidence-to-decision framework

Figure 1 traces released lifecycle results, public scenario data, exchange-ready inventories, and primary engineering evidence through gates for functional equivalence, method alignment, transformation limits, and evidence classification. The workflow returns three decision products consisting of a reproducibility result, a conditional comparison boundary, and a robustness threshold paired with the next evidence requirement. The repository records source-family metadata and a per-file path, byte count, and SHA-256 checksum. Dataset-specific tables retain the worksheet, field, unit, boundary, and numerical role used in each analysis. This is file-level integrity plus family-level provenance, not a complete flow-level data-lineage graph. The harmonization gates determine i) whether the alternatives are functionally equivalent, ii) whether the units, characterization methods, time, and geography are aligned, iii) whether the required transformation is interpolation or extrapolation, and iv) whether uncertainty, correlation, and review status are adequate for the proposed claim. The result is classified as reconstructed, screening-transformed, registered/unlinked, or supported by primary decision-grade evidence.

OpenDC-LCA is implemented in Python 3.10 or later and distributed under the MIT License [39]. The development branch evaluated here extends version 1.1.0 and provides a command-line interface, Python API, local browser GUI, scenario and performance-map schemas, reliability and replacement models, openLCA JSON-LD and Brightway mappings, evidence audits, and human- and machine-readable reports. Its comparison validator checks

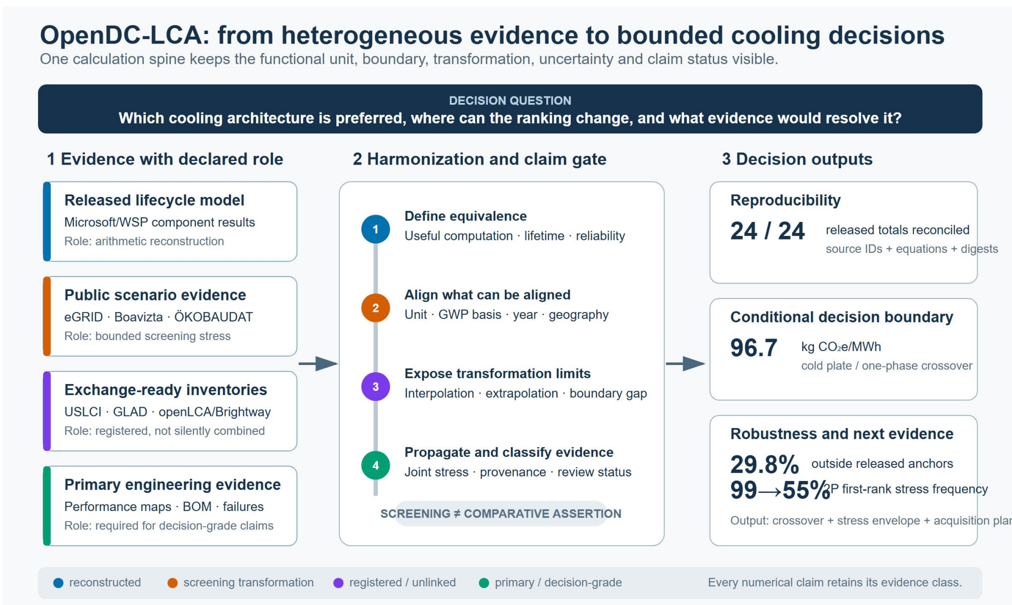

Figure 1: OpenDC-LCA evidence-to-decision architecture. Inputs are assigned a numerical role before calculation. The harmonization spine makes functional equivalence, method alignment, extrapolation, uncertainty, and claim status explicit. Outputs combine the numerical result with its evidence class and next-data requirement. The diagram layout was developed with assistance from OpenAI Codex and was reviewed and edited by the authors.

cross-scenario metadata and field-level source mappings before calculating a comparison. It cannot verify whether a declared source, uncertainty model, or review status is true. All interfaces call the same calculation engine.

2.2. Goal, functional units, and system boundaries

The practitioner package defaults to one MWh delivered to IT equipment (`it_mwh`) for operational screening. That unit is valid for comparison only when alternatives provide equivalent computational output, utilization, quality of service, reliability, and hardware life. The Microsoft/WSP secondary analysis remains in its released functional unit of one Vcore-year. No conversion between IT MWh and Vcore-year was made because the public archive does not provide a workload-independent mapping between them.

The package supports a cooling-system cradle-to-grave boundary and a broader facility cradle-to-grave boundary. The former includes cooling and heat-rejection equipment, fluids, represented maintenance and replacement, operation, and end of life. The latter can also include building, electrical, and IT assets. Common systems may be excluded only when their quantity, performance, lifetime, and end-of-life treatment are unchanged. Onsite water and electricity-supply-chain water are reported separately. Location- and market-based electricity are not merged.

2.3. Dataset roles and inclusion criteria

Table 1 distinguishes eight downloaded source families according to i) the records used, ii) their numerical role in this study, iii) the unresolved boundary or transfer limit, and iv) the permissible claim class. This source-to-claim mapping prevents software readability from being mistaken for compatibility with the case model.

Table 1: Numerical role and boundary assigned to downloaded evidence.

Source and evidence used	Numerical role in this paper	Boundary or transfer limit	Claim class
Microsoft/WSP archive with 24 totals, component tables, 2 electricity anchors, and pedigree records [14, 15]	Arithmetic reconstruction, foreground contribution model, crossover and stress analyses	Licensed background inventories and complete bills of quantities are not public	Released/ reconstructed
EPA eGRID with 459 state/ DC-year rows and 9 U.S. aggregates [40, 41]	Common-basis generation-rate scenarios and historical trend	Direct generation rates are not consumption-based lifecycle electricity inventories	Screening transforma- tion

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Table 1: Numerical role and boundary assigned to downloaded evidence. (continued)

Source and evidence used	Numerical role in this paper	Boundary or transfer limit	Claim class
Federal LCA Commons U.S. Electricity Baseline 2023 and IPCC GWP method [42]	Linked consumption-system factors for 60 balancing authorities, 10 FERC regions, and the United States	Unlinked technosphere inputs remain zero-burden cutoffs and are not harmonized with the released cooling foreground	Partial linked-system screen
Boavizta with 48 server PCF records [43]	Server-manufacturing dispersion and common-scaling stress	Products, configurations, PCRs, lifetimes, and performance are heterogeneous	Secondary cross-product evidence
ÖKOBAUDAT with 6 selected A1-A3 processes [44]	Matched steel- and cement-route procurement levers	German generic factors require architecture-specific bills of quantities	Secondary process evidence
NOAA TMY with 8,760 Fayetteville weather rows [45]	Performance-map and hourly-integration software test	No measured four-architecture cooling surface is available	Illustrative only

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Table 1: Numerical role and boundary assigned to downloaded evidence. (continued)

Source and evidence used	Numerical role in this paper	Boundary or transfer limit	Claim class
USLCI and 7 GLAD hydrogen archives [46, 47]	Exchange and interoperability tests	Product-system links, allocation, providers, and LCIA methods are unresolved	Registered/ unlinked
USGS watershed boundary with HU12 geometry [48]	Spatial data contract test	Geometry is not withdrawal, consumption, or water-scarcity characterization	Registered/ unlinked

Provider-native raw files remain outside the public repository when redistribution permission is unclear or large upstream archives have not been reviewed for redistribution. Derived tables retain filenames, worksheets, fields, units, and checksums.

2.4. Static lifecycle calculations

For IT capacity P_{IT}, utilization u, and one year, annual IT electricity is

$$E_{\mathrm{IT}} = P_{\mathrm{IT}} u (8760). \tag{1}$$

Facility electricity is

$$E_{\mathrm{facility}} = E_{\mathrm{IT}} \mathrm{PUE}. \tag{2}$$

For operational impact factor EF_k, the operational impact per delivered IT electricity is

$$I_{k,\mathrm{op}} = EF_k \mathrm{PUE}. \tag{3}$$

Equipment production and end-of-life burdens are annualized with discrete replacement according to

$$I_{k,\mathrm{eq}} = \sum_i \frac{q_i \lceil L_s/L_i \rceil \left(I_{k,i}^{\mathrm{prod}} + I_{k,i}^{\mathrm{EOL}} \right)}{L_s}, \tag{4}$$

where q_i is quantity, L_s the study period, and L_i service life. The package also supports a linear screening option. For initial fluid mass m_0, facility life L_f, annual loss m_{loss}, production factor I_k^{prod}, end-of-life factor I_k^{EOL}, and direct global-warming factor GWP_{direct},

$$m_{\mathrm{prod,annual}} = \frac{m_0}{L_f} + m_{\mathrm{loss}}, \tag{5}$$

$$I_{k,\mathrm{fluid}} = m_{\mathrm{prod,annual}} I_k^{\mathrm{prod}} + \frac{m_0}{L_f} I_k^{\mathrm{EOL}}, \qquad GHG_{\mathrm{direct}} = m_{\mathrm{loss}} GWP_{\mathrm{direct}}. \tag{6}$$

The model assumes that annual losses are replaced, so the charge remains m_0, and that the full remaining charge is treated at facility end of life. Annual purchases and outputs balance as $m_0/L_f + m_{\mathrm{loss}}$. These equations are part of the software engine. The secondary Microsoft/WSP analysis uses the normalized component values released by the authors rather than inventing missing quantities.

2.5. Released-model reconstruction

For each of four cooling architectures, two electricity cases, and three impact metrics, the released total was recomputed from the seven aggregate categories in the Nature supplementary workbook `Figure4` sheet. These categories are use phase, building, server, rack/tank, cable, supporting equipment, and fluid impacts. This 24-cell test is an arithmetic-consistency audit. It does not validate confidential quantities, licensed LCA for Experts or ecoinvent processes, allocation, performance, or field operation.

The public files do not establish one unambiguous climate-method identity for the electricity endpoints. The article reports IPCC AR5 GWP100 [14]. In the archived detailed workbook, the numeric values 524.893 and 6.133 kg CO2e/MWh appear in cells F26 labeled ‘`IPCC AR5 GTP100,” while the GWP100 cells F29 are blank. Direct worksheet-XML inspection further shows that` Comparative results 0% RE!D118 `and` Comparative results 100% RE!D122‘ reference the blank F29 cells for conventional and renewable electricity. We treat the two numbers as label-free numerical anchors, not verified GWP100 or GTP100 factors, and define an intensity-position index

$$z_s = \frac{g_s - g_R}{g_G - g_R}, \tag{7}$$

where g_G and g_R are the released GaBi electricity anchors and g_s is an external screening rate. The released impact for technology j is then transferred as

$$I_{j,s} = I_{j,R} + (I_{j,G} - I_{j,R})\, z_s. \tag{8}$$

Equations (7)-(8) reproduce both numerical endpoints exactly. They are not a background-process substitution. Common units do not resolve the endpoint method identity or the upstream-boundary mismatch with eGRID. In this study, g_s is a screening index and every extrapolation is counted. The slope of each use-phase line has units of MWh/Vcore-year and is reported as an *implied electricity-response coefficient*. It is not a measured energy demand because the two endpoint methods are unresolved.

The crossover of technologies a and b follows from equality in Eq. (8).

$$g^* = g_R + (g_G - g_R)\frac{I_{b,R} - I_{a,R}}{(I_{a,G} - I_{a,R}) - (I_{b,G} - I_{b,R})}. \tag{9}$$

2.6. eGRID harmonization

EPA changed the GWP values used by eGRID. Releases before 2018 used IPCC SAR, 2018-2022 used AR4, and 2023 used AR5 without climate-carbon feedback [40, 41]. To remove this avoidable discontinuity, every state and U.S. factor was reconstructed from net generation and gas-specific annual emissions as follows.

$$g_y^{\mathrm{AR5}} = \frac{2000M_{\mathrm{CO2},y} + 28M_{\mathrm{CH4},y} + 265M_{\mathrm{N2O},y}}{E_y}(0.45359237), \tag{10}$$

where CO2 is reported in short tons, CH4 and N2O in pounds, E_y in MWh, and the final factor converts pounds to kilograms. The provider-published CO2e rate was retained as a comparison. The provider U.S. aggregate, rather than the 50-state-plus-DC reconstruction, was used for national results. Puerto Rico was excluded from the state/DC panel. The resulting scope coverage was reported for each year.

The eGRID factors describe in-state generation. They are not consumption mixes, marginal rates, hourly signals, contractual procurement, or LCIs with upstream fuel and infrastructure. No state-specific LCA is claimed.

2.7. Federal LCA Commons electricity accounting

The 2023 U.S. Electricity Baseline was downloaded from the Federal LCA Commons as openLCA 2 JSON-LD together with the repository's IPCC GWP method [42]. The inventory contains 947 processes and 142 product systems generated with ElectricityLCI 3.0.0. We retained 71 non-residual, at-user consumption systems, consisting of 60 balancing authorities, 10 FERC regions, and the national mix. The matched residual-consumption systems were calculated separately as a market-based accounting sensitivity and were never pooled with the ordinary consumption systems.

For each product system, process links defined the technosphere coefficients. Activities were solved by fixed-point iteration for the declared target of 1 MWh at user. All exchange and target units were converted through the JSON-LD unit groups. Elementary exchanges were characterized with IPCC AR5-100 category UUID `7d05b807-2caa-3cd9-b55c-399d3b820cbc`. The calculation reports process count, link count, iteration count, maximum balance residual, impacts from generation processes, and all other linked upstream and infrastructure impacts.

An input without a product-system provider link was assigned no upstream burden, consistent with the published link structure, and was recorded as a cutoff. The national system contained 287 such exchanges. Because their omitted magnitude was not bounded and the calculation was not independently reproduced in openLCA or Brightway, the resulting factors are termed *partial linked-system factors.* They improve on a direct-generation proxy but do not constitute complete lifecycle benchmarks.

The provider scope includes trade, transmission and distribution losses, and upstream fuel supply. Generation infrastructure is included only for the provider-specified gas, oil, coal, solar, wind, and geothermal technologies. other generation infrastructure and transmission/distribution infrastructure are outside scope [42]. The very low factors in some hydropower-dominated systems must not be interpreted as complete electricity LCAs.

2.8. Transferability diagnostics

Each of the 459 state-year rates was classified as below, between, or above the two released electricity anchors. Equation (8) is interpolation only for the middle class.

The earlier analysis added transparent allowances of 0, 25, 50, 75, and 100 kg CO2e/MWh to every 2023 eGRID rate. Those values were not data-based and are retained only in the Supplementary Information as a diagnostic of the direct generation boundary. The primary boundary analysis instead uses the partial linked-system factors in Section 2.7. We also report the exact numerical clearance between the lowest 2023 direct rate and the cold-plate/single-phase crossover, $g^* - \min(g_s)$, without interpreting it as an upstream estimate.

Functional-unit sensitivity was evaluated separately. At each state-year, the minimum multiplicative increase in two-phase impact per equivalent useful computation required to equal the second-ranked architecture was

$$\delta_{\text{service}} = \frac{\min_{j \neq 2P} I_j}{I_{2P}} - 1. \tag{11}$$

This correction represents an unresolved combination of useful throughput, server count, quality of service, lifetime, and replacement. It is not a measured two-phase penalty.

2.9. Rank-robustness and stress analyses

The main robustness analysis uses equal relative half-widths so that one assumption block is not made influential merely by assigning it a wider range. For every Federal LCA Commons factor, we found the smallest half-width h at which the nominal numerical first rank could be reversed. Five sets were tested, namely i) shared grid only, ii) technology-specific use phase only, iii) embodied burden only, iv) service equivalence only, and v) all four blocks at the same h. The grid multiplier was common and nonnegative. Foreground multipliers could vary in opposite adverse directions for the nominal winner and each competitor. All monotone adverse foreground directions and both shared-grid endpoints were evaluated, and the first crossing was found by bisection. The reported threshold is an exact certificate for the declared box, not a probability or empirical uncertainty interval.

For comparison with the earlier screening analysis, we also retained seeded, 20,000-iteration triangular stress designs for the 2023 U.S. generation factor and the lowest-carbon 2023 state. The headline design used one shared grid multiplier and independent technology-specific multipliers for use phase, embodied contribution, and impact per equivalent service. It is the all-block, zero-correlation member of the dependence analysis described below. Table 2 defines the narrow, screening, and wide envelopes by pairing grid, use-phase, and service half-widths of $\pm 2\%$, $\pm 5\%$, and $\pm 10\%$ with embodied half-widths of $\pm 20\%$, $\pm 50\%$, and $\pm 100\%$, respectively.

Table 2: Declared half-widths for joint assumption-stress ensembles.

Envelope	Grid, use, and service half-width	Embodied half-width	Purpose
Narrow	±2%	±20%	Numerical stability near released values
Screening	±5%	±50%	Intermediate declared stress, not a fitted range
Wide	±10%	±100%	Deliberately severe robustness test

The triangular mode was 1.0 and the base seed was 20250730. A stable hash of the context, envelope, active block, and dependence case generated a separate seed for every design, so output does not depend on loop order. We then repeated the stress with grid only, use phase only, embodied contribution only, service equivalence only, and all four blocks active. Technology-specific multipliers were coupled with a Gaussian copula at declared latent correlations of 0, 0.5, and 1 while retaining the same triangular marginals. These correlations are not measurements. Rank frequencies describe the declared designs. They are not posterior probabilities, confidence levels, or a substitute for empirical uncertainty distributions.

Numerical convergence was checked with nested 5,000-, 20,000-, and 80,000-draw runs using the same scenario seeds. The binomial Monte Carlo standard error is reported only as integration precision within each declared design. It is not a confidence interval for cooling performance or a remedy for unsupported input distributions.

2.10. Server, construction, and data-priority analyses

Boavizta records were filtered to `Datacenter/Server` products with total GHG, manufacturing share, and lifetime. Manufacturing GHG was calculated as total product GHG times manufacturing share and annualized by declared lifetime. The candidate-flow table retains all 55 server records. Of these records, 48 met the three-field rule, while seven Lenovo records were excluded because manufacturing share was blank. No candidate was silently discarded. The minimum, quartiles, median, and maximum annualized values were divided by the empirical median and used only as common multiplicative stresses on the released compute, storage, and networking contributions. This preserves a unitless scale test without treating different servers as functionally interchangeable.

Selected ÖKOBAUDAT A1-A3 steel and cement records were compared within product families according to

$$R = 100\left(1 - \frac{I_{\text{alternative}}}{I_{\text{baseline}}}\right). \tag{12}$$

No material factor was multiplied by an assumed data-center quantity.

For data priority, released pedigree scores were averaged by component and combined with mean GHG contribution share S_c according to

$$P_c = S_c\left(\frac{D_c - 1}{4}\right), \tag{13}$$

where $D_c = 1$ is best and 5 worst. Because Eq. (13) is heuristic, rank robustness was tested using $P_c = S_c^a[(D_c - 1)/4]^b$ for $a, b \in \{0.5, 1, 2\}$. This is a screening priority, not formal value-of-information.

2.11. Temporal performance and software controls

The package can bilinearly interpolate a bounded IT-load/dry-bulb performance map and integrate its cooling-only partial PUE against hourly conditions using one declared, constant grid factor. Wet-bulb values are schema-checked but are not an interpolation coordinate, and non-cooling facility overhead is outside this partial-PUE calculation. It rejects uncovered operating points, and extrapolation is disabled. The Fayetteville TMY file and synthetic performance surface are used only in software examples because no measured four-architecture surface was available.

Reliability is represented with Weibull failure, $F(t) = 1-\exp[-(t/\eta)^{\beta}]$, renewal expectation, and optional Arrhenius life acceleration. openLCA JSON-LD and Brightway mappings preserve process and flow identifiers, providers, units, locations, and uncertainty metadata. USLCI and GLAD imports test exchange, not case-model completeness.

The scenario audit checks required metadata, duplicate identifiers, physical values, uncertainty declarations, evidence labels, and comparative intent. The comparison validator additionally requires matching functional units, system boundaries, electricity accounting, allocation and replacement rules, and LCIA methods. For a declared public assertion, each modeled input group must map to a registered source identifier. Synthetic inputs, unquantified uncertainty, and insufficient declared review remain blockers. These checks also reject non-finite numeric inputs, retain the field-to-source map in result manifests, and require a matching inclusion/exclusion record for custom comparative boundaries. They verify metadata consistency only and cannot authenticate a citation, distribution, review status, physical model, or ISO conformity [35, 36].

OpenAI Codex (OpenAI) was used in a supporting role during software development and review, test generation, plotting-code review, and manuscript organization and language editing. It did not provide primary data or serve as an evidence source. Reported numerical results were regenerated with the version-controlled deterministic workflows described above and checked against the cited sources and automated tests under author supervision.

3. Results

3.1. Released-model method identity

Figure 2 compares the released reductions relative to air cooling for cold plate, single-phase immersion, and two-phase immersion under the U.S.-grid case across GHG, primary energy, and blue water. All 24 released combinations of architecture, electricity scenario, and impact metric reconciled to the seven published Figure-4 contributions at workbook precision. Under the released U.S. grid case, the reconstructed table reports 20.66% lower GHG, 20.00% lower primary energy, and 46.78% lower blue water for two-phase immersion than air cooling. These are reproduced source results, not independent comparisons.

The cell-level audit changed their permissible use. The article identifies AR5 GWP100, but the detailed workbook's numeric endpoint cells are labeled AR5 GTP100 and the corresponding GWP100 cells are blank. Comparative use-phase formulas also reference those blank F29 cells. The two numeric endpoints lack a resolved LCIA-method identity in the public archive. The implied use-phase response coefficients were 0.06317, 0.05364, 0.05259, and 0.05006 MWh/Vcore-year for air, cold plate, single-phase, and two-phase, respectively, with zero-intensity intercepts of 0.010-0.013 kg CO2e/Vcore-year. These slopes are useful diagnostics, but they are not independently measured electricity demand.

3.2. Historical generation-emission intensity

The common-basis U.S. generation factor declined from 517.731 kg CO2e/MWh in 2012 to 349.667 kg CO2e/MWh in 2023, a 32.462% reduc-

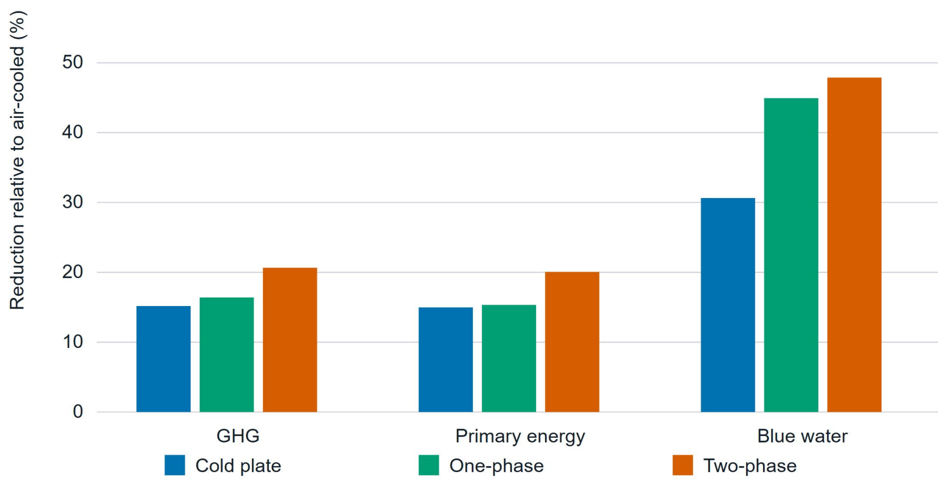

Figure 2: Relative GHG, primary-energy, and blue-water reductions for cold plate, single-phase immersion, and two-phase immersion versus air cooling under the released U.S.-grid case. The 24 architecture, electricity, and impact combinations reconcile at workbook precision and reproduce the released arithmetic without validating the underlying inventories, LCIA-method identity, or field performance.

tion. The trajectory was not monotonic. The factor rose from 373.097 in 2020 to 388.699 in 2021 before declining. Reconstructing every year with AR5 GWP100 changed the provider-published national rate by no more than 0.248 kg CO2e/MWh, in 2012. The direct CO2-only rate was 348.000 kg/MWh in 2023. Thus, the long-run decline is not an artifact of eGRID's transition from SAR to AR4 and AR5.

The 50-state-plus-DC reconstruction matched the official national result through 2018. From 2019 onward it was 0.32-0.43% lower because the panel covered 99.55-99.58% of the provider U.S. generation after excluding Puerto Rico. Figure 3 places all 459 harmonized state/DC generation rates from nine eGRID releases on a common kg CO2e/MWh axis, overlays the reported provider U.S. aggregate trajectory as an orange polyline, and identifies points below the 96.704 kg CO2e/MWh cold-plate/single-phase crossover in blue. The provider aggregate was retained for the national series and was not fitted to the state/DC points. Across the full panel, 137 of 459 numerical rates lay outside the two released cooling anchors, so 29.85% of the transfers were extrapolations.

3.3. Lifecycle electricity factors

Figure 4 compares the partial linked-system GHG intensity of 10 FERC-region consumption systems and the national system on a delivered-electricity basis. Each bar separates generation-process impacts from other linked upstream and infrastructure impacts, while the two dashed lines locate the 96.704 and 524.893 kg CO2e/MWh released numerical anchors.

The 71 non-residual Federal LCA Commons systems ranged from 0.921 to 987.844 kg CO2e/MWh. The national linked-system factor was 422.931 kg

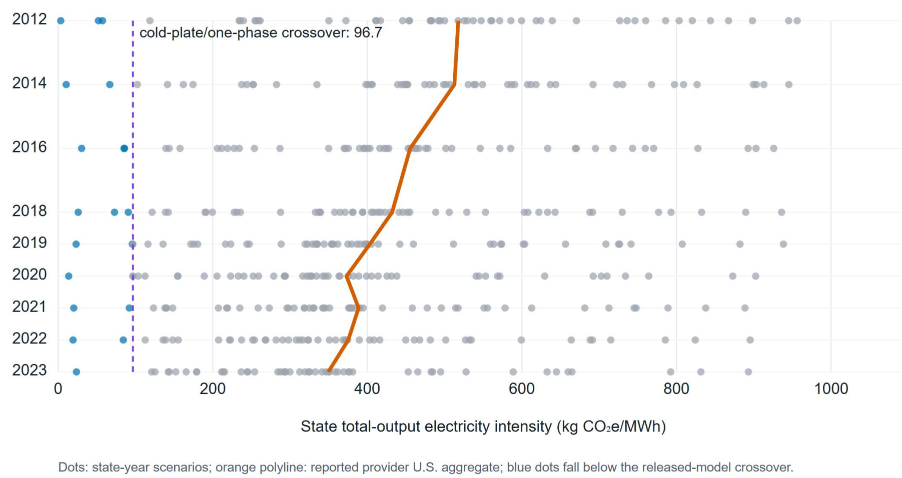

Figure 3: Harmonized AR5-GWP100 eGRID state/DC generation rates for nine releases, with electricity intensity on the horizontal axis in kg CO2e/MWh. The orange polyline connects the reported provider U.S. aggregate for each year. Blue points fall below the released-model numerical cold-plate/single-phase crossover. These are direct-generation scenarios, not consumption LCIs or state cooling LCAs.

CO2e/MWh. It comprised 369.848 from electricity-generation processes and 53.084 from other linked upstream and infrastructure processes. It exceeded the 2023 national eGRID direct-generation rate by 73.264 kg CO2e/MWh, although the difference also contains consumption-mix, delivery-loss, flow-coverage, and boundary effects and is not an isolated upstream correction.

All 10 FERC-region systems exceeded the 96.704 kg CO2e/MWh numerical cold-plate/single-phase crossover. Their factors ranged from 238.897 kg CO2e/MWh for NYISO to 557.110 kg CO2e/MWh for MISO. Ten of 60 balancing-authority systems fell below the crossover, and 15 of all 71 systems exceeded the released high anchor of 524.893 kg CO2e/MWh. The numerical cooling screen placed two-phase first in all 71 systems, but this is not a common-method comparative LCA because the cooling foreground method remains unresolved.

The national result converged in five fixed-point iterations with a balance residual of zero at the reporting precision, using 582 processes and 1,174 links. It also contained 287 unlinked technosphere exchanges across 47 processes, including fuels, construction materials, chemicals, and reclaimed water. Their upstream burdens were zero in the linked calculation and were not bounded. The factor is more complete than eGRID direct emissions but still a partial linked-system result.

The national residual-consumption system was 455.350 kg CO2e/MWh, comprising 399.934 from generation processes and 55.417 from other linked processes. It was 7.67% above the ordinary consumption-system result. This difference is an electricity-accounting sensitivity, not an uncertainty interval, and the two accounting products were not mixed within any cooling screen.

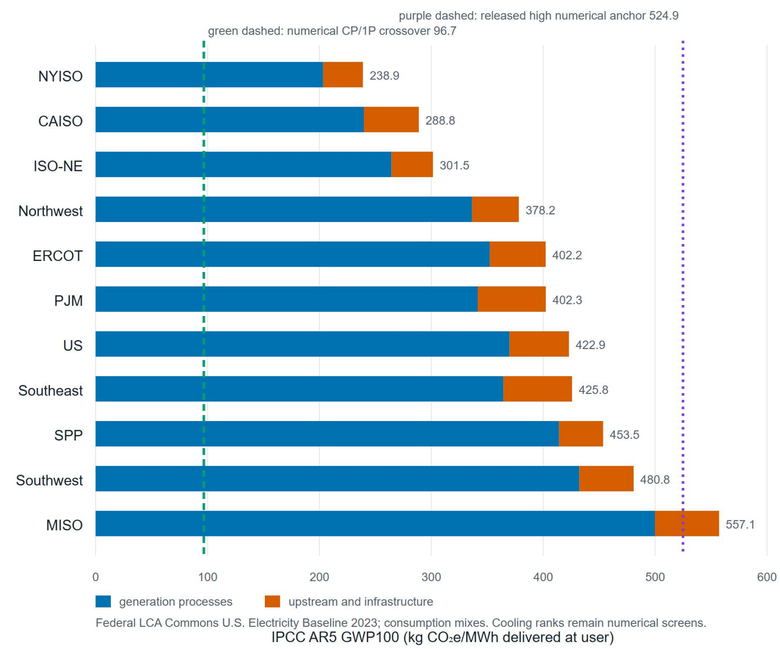

Figure 4: IPCC AR5-GWP100 partial linked-system electricity factors for 10 FERC regions and the United States. The horizontal axis is kg CO2e/MWh delivered at the user. Bars separate generation-process impacts from other linked upstream and infrastructure impacts. The green dashed line marks the 96.704 kg CO2e/MWh cold-plate/single-phase numerical crossover. The purple dashed line marks the 524.893 kg CO2e/MWh released high numerical anchor. Neither line creates method equivalence.

3.4. Rank-robustness thresholds

Figure 5 compares the national critical half-widths for shared grid, use phase, embodied burden, service equivalence, and simultaneous equal-width variation in panel A. Panel B plots the all-block critical half-width against lifecycle electricity intensity for 60 balancing authorities, 10 FERC regions, and the national system.

At the national lifecycle factor, the numerical two-phase first rank could not be reversed by a shared grid-factor change within the tested nonnegative range through a 99.999% half-width. With technology-specific variation, reversal occurred at 2.994% for use phase, 22.154% for embodied burden, and 2.637% for service equivalence. When all four blocks used the same half-width, the first reversal occurred at 1.318%, against single-phase.

Across all 71 electricity contexts, the all-block critical half-width ranged from 1.123% at PUD No. 1 of Douglas County to 1.479% at Duke Energy Progress West. At the lowest-carbon system, the critical competitor was cold plate. At that point, the embodied-only and service-only thresholds were 2.291% and 2.246%, while use-phase and shared-grid changes did not reverse rank through 99.999%. The ranking is numerically fragile even though it survives every unperturbed electricity factor.

This equalized analysis reverses one conclusion from the earlier triangular stress design. Embodied variation had appeared to cause the largest rank loss because its assigned half-width was 10 times those of use phase and service. At equal widths, embodied burden has the largest national one-block tolerance, whereas service equivalence and use phase are limiting. The earlier stress frequencies and their dependence/convergence diagnostics remain in the Supplementary Information as evidence about the chosen design, not the technologies.

The one-sided functional-unit diagnostic gives a complementary threshold. Across 2023 direct-rate contexts, a 5.24-6.04% adverse correction to two-phase impact per equivalent useful computation would erase its numerical first rank. This is a break-even requirement for future measurements, not an observed performance difference.

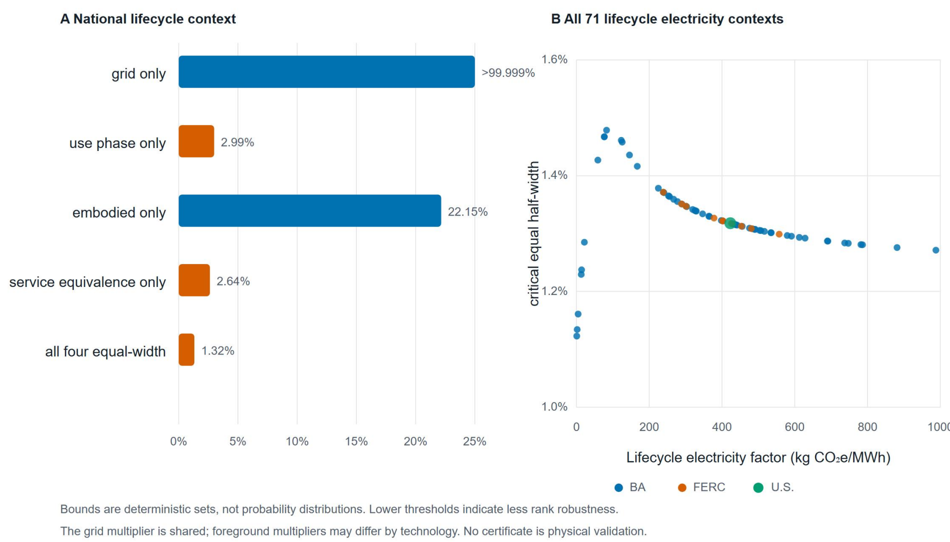

Figure 5: Deterministic equal-width rank-robustness certificates. (A) National critical half-width by active block. (B) all-block thresholds across 60 balancing authorities, 10 FERC regions, and the national system. Bounds are uncertainty sets, not probability distributions or physical validation.

3.5. Server and construction evidence

Of 55 Boavizta `Datacenter/Server` candidates, 48 included records spanned 465-2,503 kg CO2e/server for manufacturing, a 5.38-fold range. Annualized values spanned 116-626 kg CO2e/server-year. Commonly scaling the released server-related contributions across the empirical 0.38-2.06-fold annualized envelope did not change the deterministic two-phase first rank. That structural stress does not model architecture-specific server count, configuration, throughput, or lifetime.

The selected ÖKOBAUDAT high-scrap electric-arc-furnace galvanized steel record had 59.9% lower A1-A3 GHG, 47.9% lower nonrenewable primary energy, and 52.0% lower freshwater use per kilogram than the selected low-scrap blast-furnace-route record. The selected CEM III cement record had 49.8% lower GHG, 12.1% lower nonrenewable primary energy, and 23.5% lower freshwater use than CEM II/A. These are procurement levers, not data-center reductions. Their facility effect remains unknown until each architecture has a reviewed bill of quantities.

3.6. Data-priority sensitivity

Under the base contribution-times-weakness index, use phase was the highest priority, followed by storage, networking, and compute-server evidence. Across the nine alternative exponent combinations, use phase ranked first in eight and in the top three in eight. Networking ranked first once, storage remained in the top three in all nine, and compute was in the top three in four. The rank of use phase ranged from first to fourth.

This sensitivity changes the interpretation of the pedigree result. It supports a near-term measurement sequence—hourly IT and cooling electricity, then server inventories and functional performance—but not a unique value of information. A formal acquisition decision requires empirical parameter distributions, correlations, measurement cost, and the loss associated with a wrong architecture choice [34].

3.7. Software verification and practitioner outputs

The full local test suite covers calculation, fluid mass conservation, interoperability, reliability, public-data parsing, practitioner workflow, GUI/API parity, endpoint reconstruction, eGRID harmonization, Federal LCA Commons process-system solution, cutoff reporting, equal-width robustness, comparison incompatibility, field-level lineage, stress-design determinism, and provenance integrity. Paper-reconstruction tests skip explicitly when locally held provider files are absent. No substitute values are inserted.

A practitioner supplies IT capacity, utilization, PUE, facility lifetime, onsite water, location- and market-based electricity factors, optional equipment/fluid inventories, and a citable source record. The package returns annual and per-IT-MWh impacts, contribution breakdowns, evidence classification, warnings or blockers, a scenario digest, model version, and the next-data list. Advanced users can add hourly performance maps, replacement and reliability, and openLCA/Brightway inventories. Software consistency does not validate user inputs, the truth of metadata labels, or the physical adequacy of a screening model.

4. Discussion

4.1. Comparative claim boundaries

The released table and every unperturbed electricity screen place two-phase first. Three independent findings prevent that order from becoming a public comparative conclusion. First, the archive does not resolve the climate method of the numeric electricity endpoints. Second, the cooling foreground cannot be recalculated under the same AR5 method as the electricity inventories. Third, useful computation has not been measured across the four architectures. The most important outcome is the boundary between reproduced arithmetic and a physically identified comparison.

The 96.704 kg CO2e/MWh cold-plate/single-phase crossover is a property of two released numerical endpoints. It helps locate where their affine lines cross, but it is not an environmental threshold for a real site. Likewise, survival across 71 electricity factors explores one background dimension. It does not provide 71 independent observations of cooling performance. A transferable assessment must report the numerical result together with its functional unit, method identity, system boundary, cutoff structure, robustness definition, and claim status.

4.2. Electricity-system boundaries

The eGRID reconstruction shows that the 32.46% historical national decline is robust to changing GWP conventions. The Federal LCA Commons calculation then replaces an arbitrary additive boundary allowance with a provider-linked consumption model. At the national point, linked upstream and infrastructure processes add 53.084 kg CO2e/MWh beyond generation processes, and the total is 73.264 kg CO2e/MWh above the eGRID national rate. All FERC regions lie above the numerical cold-plate/single-phase crossover.

Provider-linked consumption modeling improves boundary traceability but does not complete harmonization because the difference between the two national factors also reflects delivery losses, consumption versus generation geography, and elementary-flow coverage. More importantly, 287 unlinked technosphere exchanges still carry zero upstream burden. A complete calculation requires those providers to be linked or their omitted impacts to be bounded and independently reproduced. Even then, electricity cannot repair the unresolved cooling-foreground LCIA method. Location- and market-based cases must also remain consistent throughout the product system [22, 23]. The 7.67% national difference between ordinary and residual consumption mixes shows that the accounting choice is consequential even before the cooling foreground is harmonized.

4.3. Robustness-bound design

The earlier triangular analysis assigned embodied burden a half-width 10 times the use-phase and service widths. Its conclusion that embodied variation was the largest one-block source of rank loss conflated sensitivity with an arbitrary range choice. The equal-width certificate separates range assignment from model sensitivity. At the national lifecycle factor, embodied burden can vary by 22.15% before a worst-case one-block reversal, whereas service equivalence and use phase require only 2.64% and 2.99%.

The all-block threshold of 1.32% is not a confidence bound. It is the radius of a specified box in which a shared grid multiplier and adverse technology-specific foreground multipliers cannot reverse the numerical first rank. The certificate is falsifiable because any proposed empirical uncertainty model can be compared against the same bound structure. A probability of rank would additionally require measured marginal distributions and correlations [30].

Because functional equivalence and use-phase demand are limiting at equal widths, near-term measurements should close those two evidence gaps first, while bills of quantities and material inventories remain necessary for a common-method lifecycle comparison. The legacy triangular results still demonstrate that dependence assumptions matter, but they do not rank the real-world importance of uncertainty blocks.

4.4. Functional equivalence

Vcore-year is more informative than facility area or rack count, but it is still a proxy. Equivalent useful computation depends on workload, accelerator utilization, memory and network constraints, throttling, overclocking, availability, and quality of service. The one-sided 5.50% median threshold and the symmetric 2.64% national service bound are break-even requirements, not measured penalties. They define the approximate resolution required of a comparative experiment. ITU-T L.1410 similarly requires an explicit functional unit and comparative baseline for ICT services [18].

A shared protocol should collect several forms of evidence for every architecture. It should measure i) completed workload or benchmark output, ii) IT and cooling power, and iii) inlet and component temperatures together with flow and pressure drop. It should also record iv) fan, pump, cooling distribution unit, and heat-rejection power, v) onsite water use and failure/maintenance events, and vi) uncertainty and calibration lineage over the same load-weather domain. Server count, configuration, and replacement must be tracked with that performance. A thermal result without useful computation cannot close the LCA functional unit. A product carbon footprint without architecture quantity cannot close the embodied inventory.

4.5. Grid decarbonization and evidence priorities

National grid decarbonization simultaneously lowers every electricity-dependent cooling result, contracts the absolute value of operational efficiency, and increases the fraction supplied by embodied terms. The absolute modeled two-phase-versus-air benefit fell 29.45% between the 2012 and 2023 generation conditions, a change that percentage savings alone do not reveal.

This shift explains why the Boavizta and ÖKOBAUDAT datasets matter even though they were not inserted into architecture totals. The 5.38-fold server manufacturing range is larger than many cooling differences, but heterogeneous PCFs cannot be treated as an uncertainty distribution for interchangeable servers. The approximately 50-60% material-route GHG levers are large per kilogram, but kilograms by architecture are missing. The needed research product is a linked foreground dataset that combines useful computation, server configuration and lifetime, cooling bill of quantities, material route, and measured operation under one functional unit.

4.6. Relationship to LCA software

OpenDC-LCA does not replace openLCA, Brightway, Temporalis, premise, or bw_timex. Those tools provide process-network calculation, database management, prospective backgrounds, or time-explicit LCA [27, 28, 29, 37, 38]. The OpenDC-LCA assessment layer adds six comparison-audit functions. First, it checks functional equivalence against useful computation. Second, it represents load-weather cooling performance and performs bounded annual integration. Third, it represents reliability, replacement, and cooling fluids. Fourth, it separates lifecycle electricity, direct generation rates, market instruments, and marginal signals. Fifth, it preserves source and transformation provenance. Finally, its automated metadata validator blocks declared comparisons with incompatible methods, boundaries, or source mappings.

The metadata validator is deliberately narrower than a scientific review because it trusts the metadata supplied by the user and cannot determine whether a performance map is representative, a distribution is empirical, or an external review occurred. Its role is to catch machine-detectable incompatibilities before calculation, not to authorize a public environmental claim.

This comparison-audit role also differs from recent data-center performance and architecture models. AlphaDataCenterCooling provides operational validation of a cooling plant [17], while d'Orgeval et al. compare full data-center architectures and, for a subset, normalize by computational performance [16]. The present work tests whether published secondary LCA results retain their meaning when moved across electricity datasets and claim contexts. It does not duplicate either primary modeling capability.

This is why USLCI and GLAD hydrogen archives were exchange-tested but not forced into the results. Backup power is a relevant future application, yet a hydrogen dataset becomes numerical evidence only after technology, electricity source, compression/storage, allocation, geography, and LCIA method are linked to a declared product system.

4.7. Hyperscale applications

The most useful extension of the Microsoft/WSP work is not another static national scenario. It is a jointly governed evidence package that allows the released model to be updated without weakening its functional-unit insight. Closing the method, foreground, performance, electricity, and uncertainty gaps requires five additions. First, the original GWP100 electricity factors or elementary-flow inventories are needed to resolve the public workbook's blank F29 cells. Second, the package requires architecture-specific, anonymized bills of quantities and service lives. Third, it requires measured performance surfaces for IT output, server power, facility parasitics, and on-site water across load and weather. Fourth, it requires a complete provider-linked lifecycle electricity model under the same LCIA method, with cutoffs linked or bounded and with location-based, market-based, and marginal operational signals reported separately. Finally, empirical uncertainty and correlation information are needed for rank-probability and value-of-information analyses.

OpenDC-LCA supplies schemas, transformations, tests, and claim boundaries for that collaboration. A preliminary university high-performance-computing site characterization illustrates why those claim boundaries matter in practice. The site combines dedicated IT and liquid-cooling paths with raised-floor CRAC cooling shared by several institutional users. It therefore cannot support a single facility PUE without a documented allocation of shared support loads. The preliminary record contains no operational time series and does not report site energy, cooling performance, PUE, or environmental impacts. It instead defines the metering sequence required to produce those results.

The custom-boundary validator now classifies each energy item as metered, allocated, or excluded. Metered records identify their meter, allocated records identify an allocation method, and excluded records state a rationale. A facility-PUE label requires a complete energy-item inventory, a metered IT-load boundary, and no excluded shared-support item. A partial record can instead report a dedicated-support ratio for the stated UPS and dedicated-cooling boundary. This check does not authenticate meter data or determine whether an allocation is physically representative, but it prevents incomplete shared infrastructure from being silently labeled as facility energy.

The two conclusions supported independently here are the released arithmetic and the national direct-generation trend. The cooling crossovers, first ranks, and response coefficients remain conditional numerical screens until the five evidence gaps are closed.

4.8. Limitations and evidence requirements

Because this analysis is secondary, it does not recreate proprietary background inventories, confidential bills of quantities, the full virtual-core model, or measured hyperscale operation. Equation (8) uses a numerical axis whose climate-method identity is unresolved in the public archive. The Federal LCA Commons factors contain 287 national unlinked technosphere exchanges. The custom solver has not been independently reproduced in openLCA or Brightway. The additive allowance, triangular marginals, and copula dependence cases are secondary diagnostics, not estimates. The convergence check limits numerical sampling error within those designs. It does not reduce epistemic uncertainty in the ranges, dependence structure, cutoffs, or foreground model. eGRID state rates omit consumption transfers, contracts, hourly dispatch, and marginal effects.

The climate-focused extension does not provide a boundary-consistent comparison of primary energy, water scarcity, toxicity, resource use, or other impact categories. The Boavizta sample mixes products and methods, while ÖKOBAUDAT records are German generic A1-A3 factors. The TMY and synthetic performance examples validate software pathways, not technology performance. The USGS file provides watershed geometry rather than water consumption or AWARE characterization. Reliability models omit dependent failures, repair queues, redundancy and maintenance logistics unless the user supplies them. A coauthor methodology review is not an independent critical review.

The evidence supports a methodological Technology Assessment of transferability, not a public assertion that one cooling architecture is environmentally preferable. Such an assertion would require resolved common-method inventories, reviewed primary performance and useful-computation data, architecture bills of quantities, water-scarcity characterization, empirically supported joint uncertainty, and an external critical review panel.

5. Conclusions

This assessment separates three evidence levels that had previously been combined. These levels are reproduced source arithmetic, direct-generation trend analysis, and conditional cooling screens. All 24 published totals were arithmetically reproduced, but the public archive does not resolve whether the two numeric electricity endpoints used by the screening axis are AR5 GWP100 or GTP100. Cooling ranks derived from that axis cannot be interpreted as common-method comparative LCA results.

The national eGRID generation factor declined 32.46% from 2012 to 2023 after all nine releases were harmonized to AR5 GWP100. The Federal LCA Commons national consumption system yielded 422.93 kg CO_2e/MWh, including 53.08 kg CO_2e/MWh from linked processes beyond generation, and all 10 FERC regions lay above the 96.70 kg CO_2e/MWh numerical crossover. These partial linked-system factors retain documented technosphere cutoffs and require independent calculator verification before benchmark use.

Equal-width robustness analysis corrected the earlier attribution of rank loss to embodied variation. At the national lifecycle factor, the first-rank screen failed at 2.64% service-equivalence variation, 2.99% use-phase variation, and 22.15% embodied variation. Simultaneous equal-width changes failed at 1.32%. The dominant near-term evidence needs are architecture-resolved useful computation and use-phase demand, followed by complete bills of quantities and common-method inventories.

The OpenDC-LCA validator rejects configured comparisons whose declared metadata contain mismatched functional units, boundaries, electricity accounting, LCIA methods, or missing field-level source mappings. This check detects declared incompatibilities but does not verify the underlying science. The study is suitable as a transferability Technology Assessment. It does not support a public claim of environmental superiority among cooling architectures.

6. Data and code availability

OpenDC-LCA version 1.1.0 is publicly available at `https://github.com/UARK-NED3/OpenDC-LCA` and `https://github.com/UARK-NED3/OpenDC-LCA/releases/tag/v1.1.0`. The development revision evaluated here adds the Federal LCA Commons solver, cutoff reporting, equal-width robustness certificates, and comparison-level metadata checks. It will be tagged and archived before submission. Analysis metadata and the provider-file manifest record file sizes and SHA-256 checksums. Provider-native files remain outside Git when redistribution permission is unclear or a large upstream archive has not been reviewed for redistribution. The Microsoft archive, eGRID files, Federal LCA Commons baseline and method, and NOAA data are available from their public providers [15, 40, 41, 42, 45]. No DOI-bearing software archive is available at the time of this draft.

7. Competing interests

Braden Stevens and Pengjiang Xiang are full-time employees of Harrison French & Associates Ltd.

8. Ethics statement

This secondary study used public or author-held engineering datasets and did not involve human participants or animals. Research-ethics approval was not required.

9. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the authors used OpenAI Codex to support code review, test development, figure-code review, manuscript organization, and language editing. The authors reviewed and edited the resulting material, verified the reported calculations and cited sources, and take full responsibility for the content of the article.

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